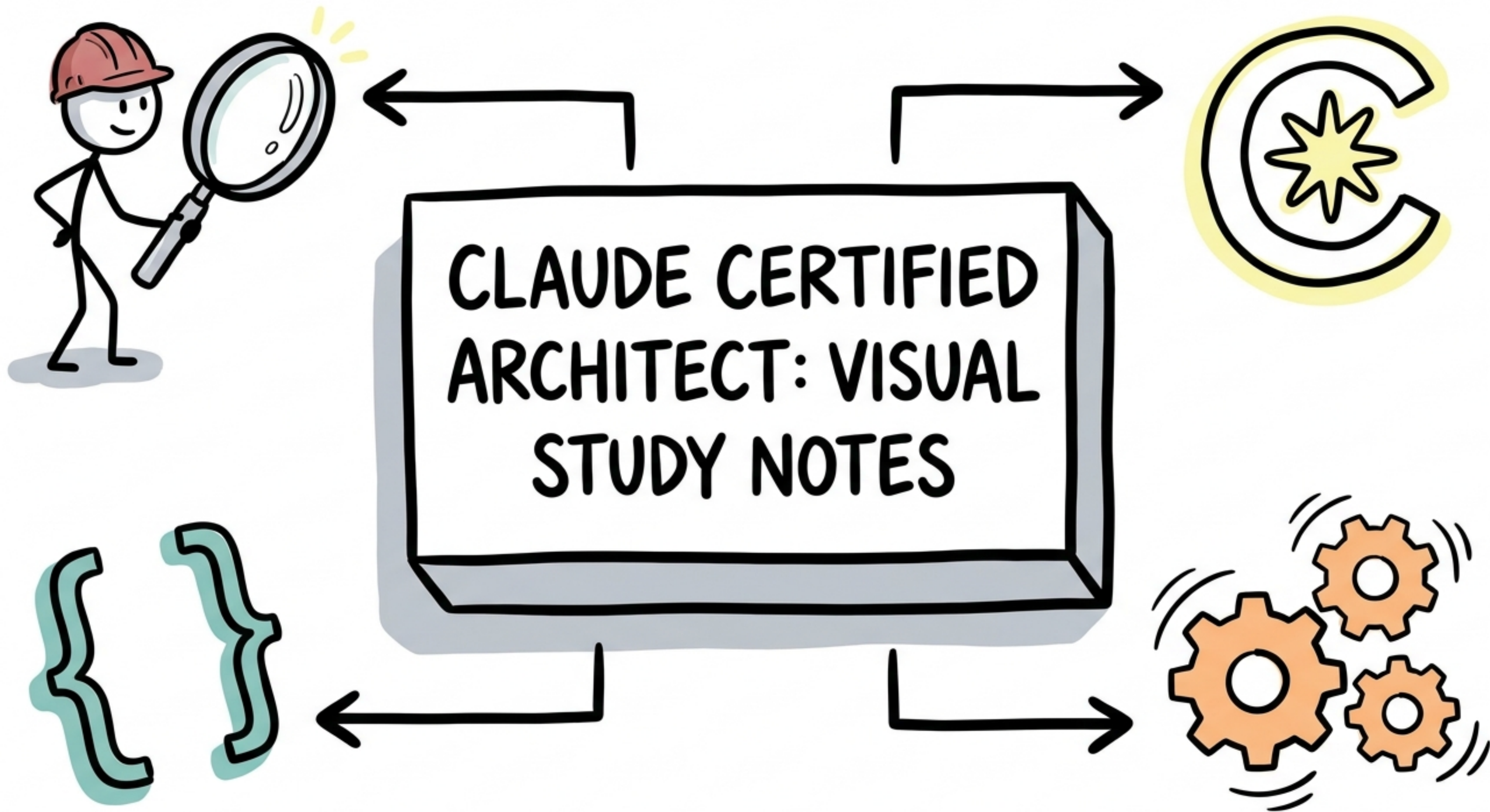




THE CCAR-F BLUEPRINT MATRIX

DOMAIN 5:
CONTEXT MANAGEMENT
& RELIABILITY



DOMAIN 4:
PROMPT ENGINEERING
& STRUCTURED OUTPUT



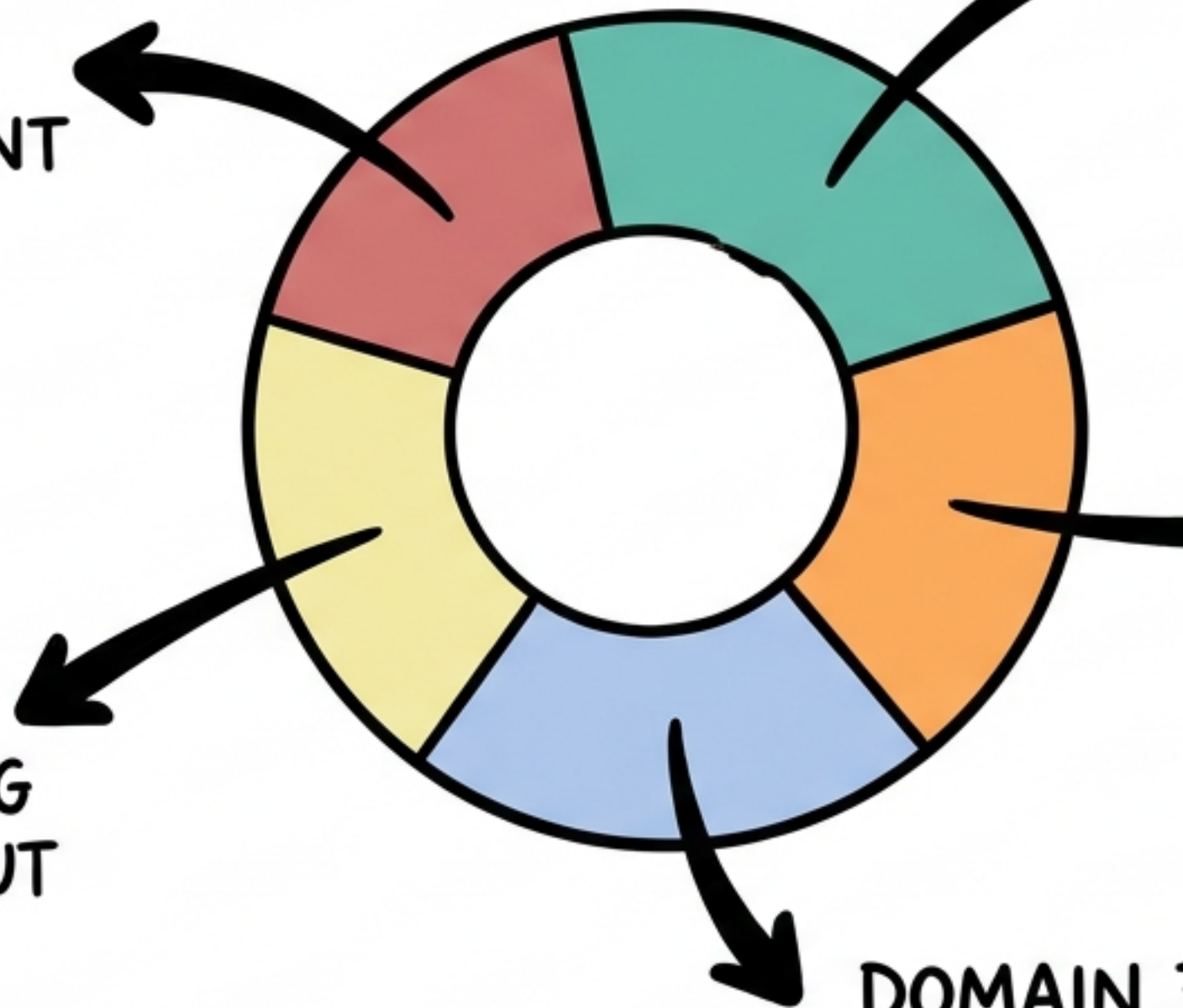
DOMAIN 1:
AGENTIC ARCHITECTURE
& ORCHESTRATION



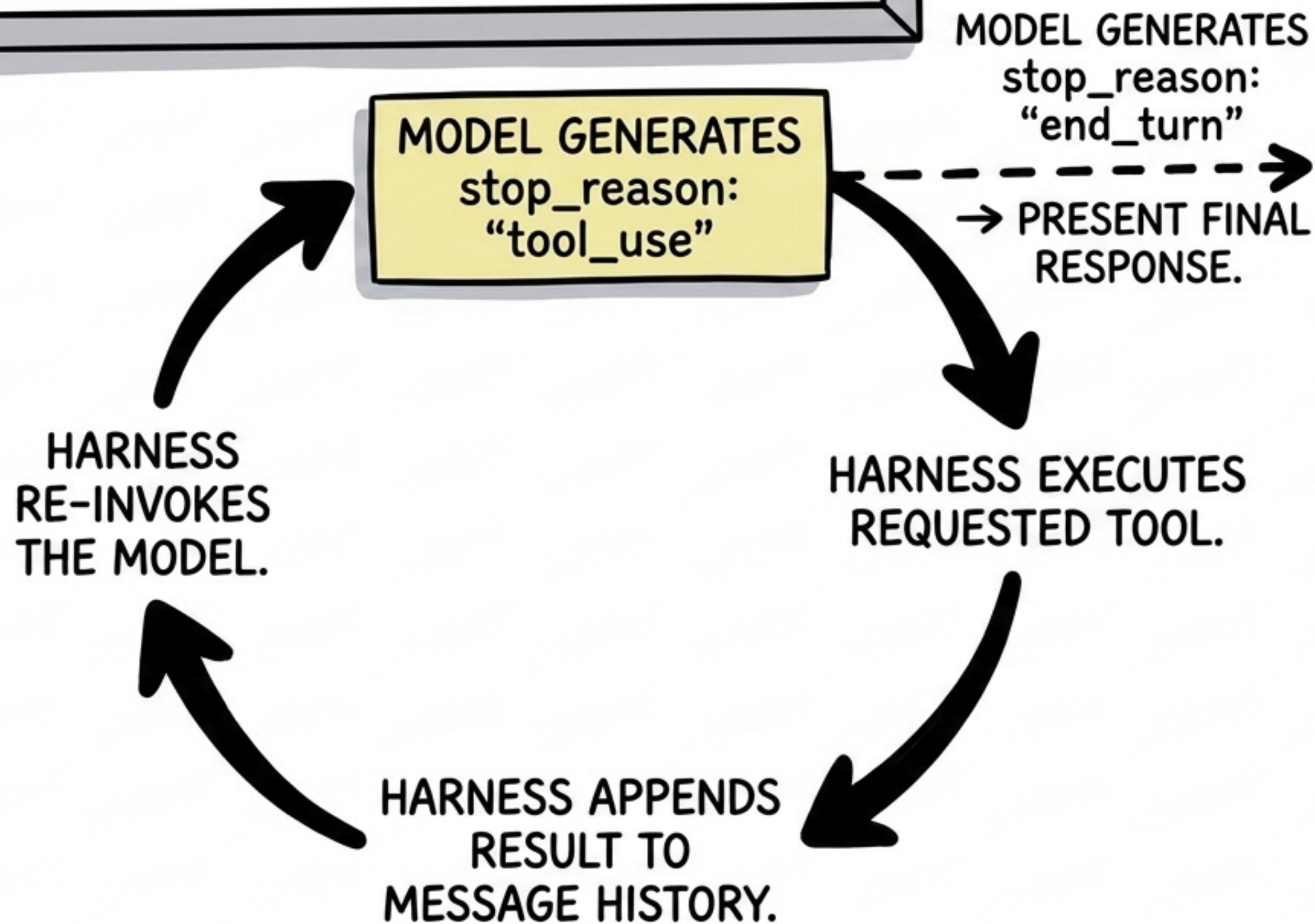
DOMAIN 2:
TOOL DESIGN &
MCP INTEGRATION



DOMAIN 3:
CLAUDE CODE CONFIG
& WORKFLOWS



D1: THE CORE AGENTIC LOOP

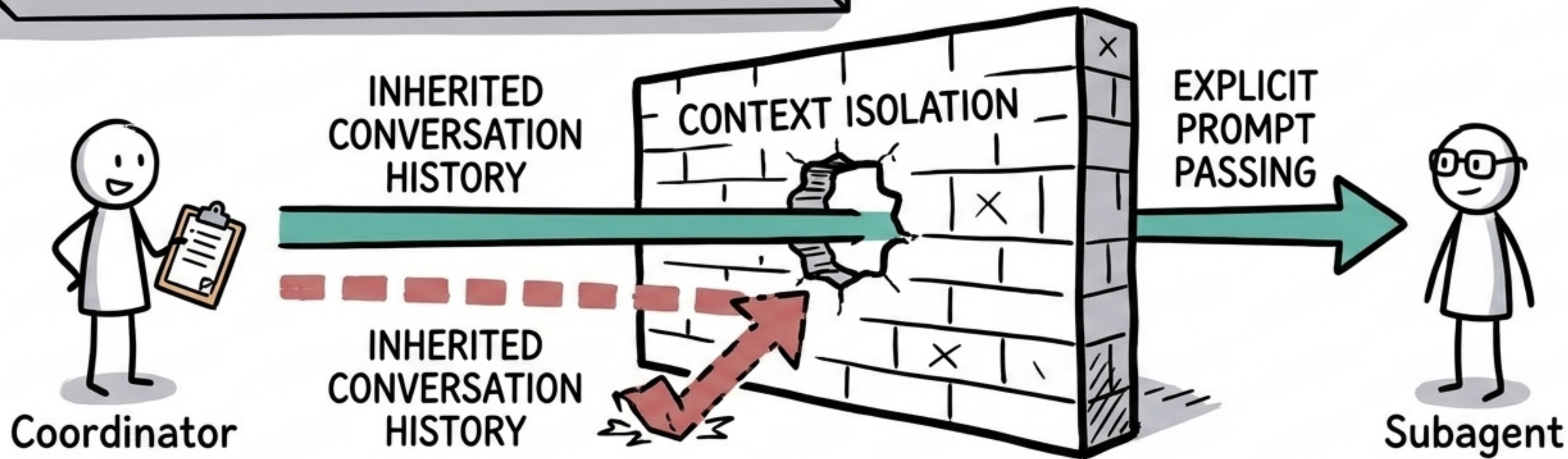


WAIT FOR THE
ASSISTANT TO SAY
"I HAVE RESOLVED
THE ISSUE". ∴



NEVER PARSE
NATURAL LANGUAGE
FOR COMPLETION
SIGNALS.
RELY ENTIRELY ON
API STOP REASONS.

D1: ORCHESTRATING SUBAGENTS



DO EXACTLY
THESE
5 STEPS



GOALS &
SUCCESS
CRITERIA



PRO-TIP: NEED PARALLEL EXECUTION?
EMIT MULTIPLE 'TASK' TOOL CALLS IN A SINGLE RESPONSE.

D2: DESIGNING FAILURE STATES

OPERATION FAILED.



MODEL HAS NO SIGNAL
FOR RECOVERY.



```
{  
  errorCategory: 'business'  
  isRetryable: false  
  description: 'Violates  
    30-day return policy.'  
}
```

- USE MCP'S 'ISERROR' FLAG FOR TRANSPORT/PARSING FAILURES.
- POLICY VIOLATIONS ARE BUSINESS ERRORS (NON-RETRYABLE). LET THE AGENT EXPLAIN THEM TO THE USER.

D2: BUILT-IN TOOLS & MCP CATALOGS

GLOB

FINDS PATHS
(e.g., `**/*.test.tsx`)

GREP

FINDS CONTENTS
(e.g., `'PaymentDeclinedError'`)

EDIT

FAILS IF
ANCHOR TEXT
ISN'T UNIQUE

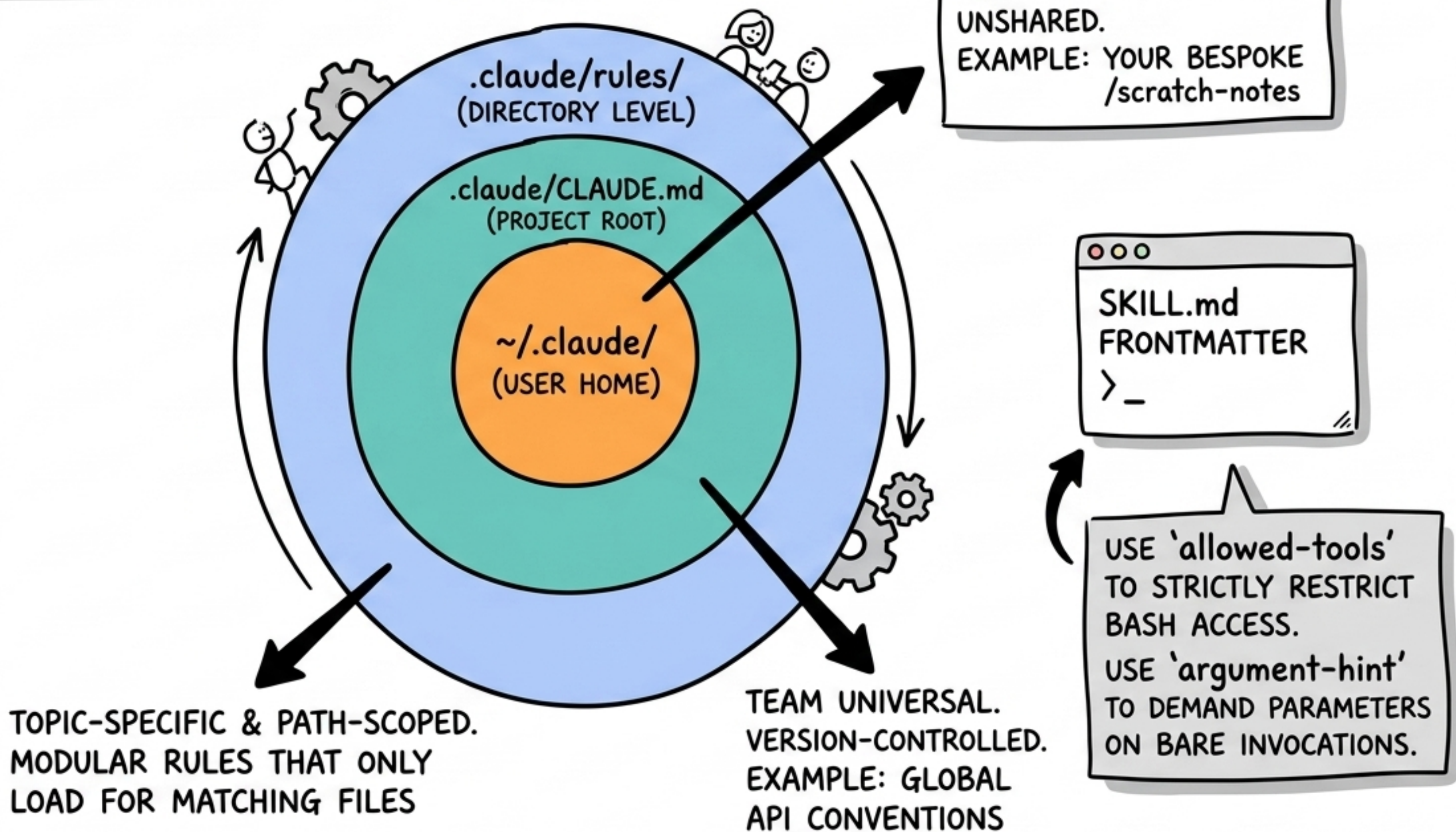
READ + WRITE

THE FALLBACK
FOR REPETITIVE
LEGACY FILES



DON'T BURN TOOL CALLS EXPLORING.
EXPOSE CONTENT CATALOGS (SCHEMAS,
DOC HIERARCHIES) AS MCP RESOURCES
SO AGENTS START INFORMED.

D3: CLAUDE CODE CONFIG SCOPING



D3: CI/CD & SESSION STATE

fork_session

COMPARE COMPETING
REFACTORS FROM A SHARED
ANALYSIS BASELINE WITHOUT
CROSS-CONTAMINATION.



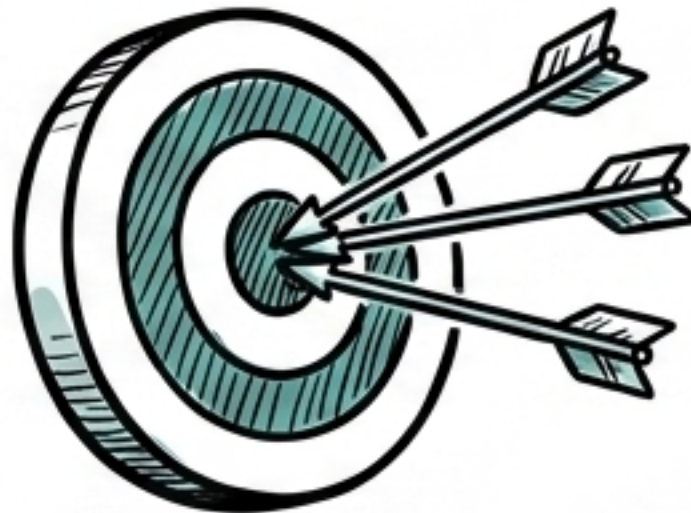
SESSIONS KEEP THE REASONING THAT PRODUCED
THE CODE, BIASING REVIEWS. ALWAYS USE AN
INDEPENDENT, HEADLESS RUN FOR CI REVIEWS.

```
>_ claude -p --output-format json --json-schema
```

D4: PRECISION & FEW-SHOT REVIEWS



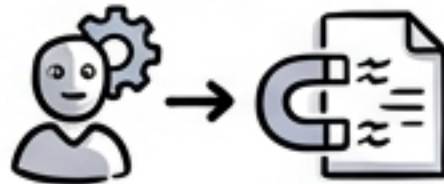
FEW-SHOT ENGINEERING



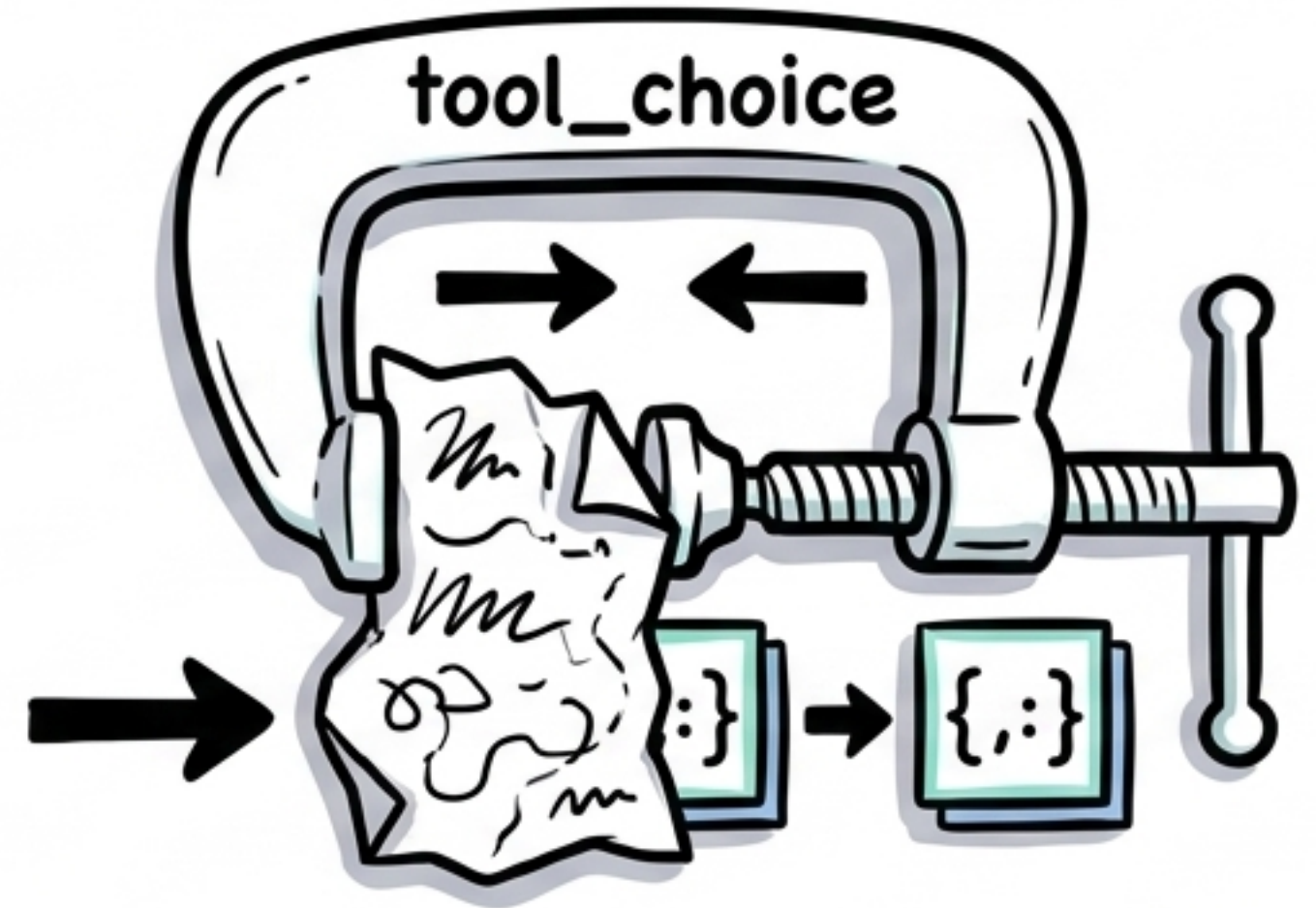
HOW FEW-SHOT WORKS BEST:
DON'T USE 20 GENERIC EXAMPLES.
USE 2-4 TARGETED EXAMPLES AIMED
AT AMBIGUOUS SCENARIOS, SHOWING
THE REASONING FOR WHY ONE ACTION
BEATS ANOTHER.

D4: IRONCLAD STRUCTURED OUTPUT

1. FORCING THE SHAPE:



USE **'tool_choice: any'** FOR UNKNOWN DOCS.
MODEL MUST CALL AN EXTRACTION TOOL,
NEVER OUTPUTTING CONVERSATIONAL TEXT.



2. PREVENTING HALLUCINATIONS:

IF A FIELD (LIKE A TAX ID)
MIGHT BE MISSING, MAKE IT
NULLABLE / OPTIONAL IN
THE SCHEMA.

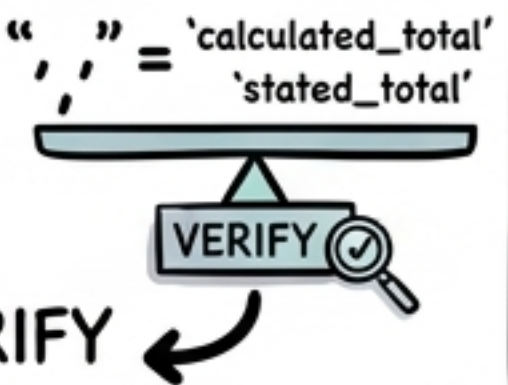


3. STRUCTURE VS SEMANTICS:

SCHEMAS GUARANTEE **SYNTAX**
(COMMAS/QUOTES).

THEY DON'T GUARANTEE **MATH**.

ADD DOWNSTREAM LOGIC TO VERIFY
'calculated_total' VS 'stated_total'.



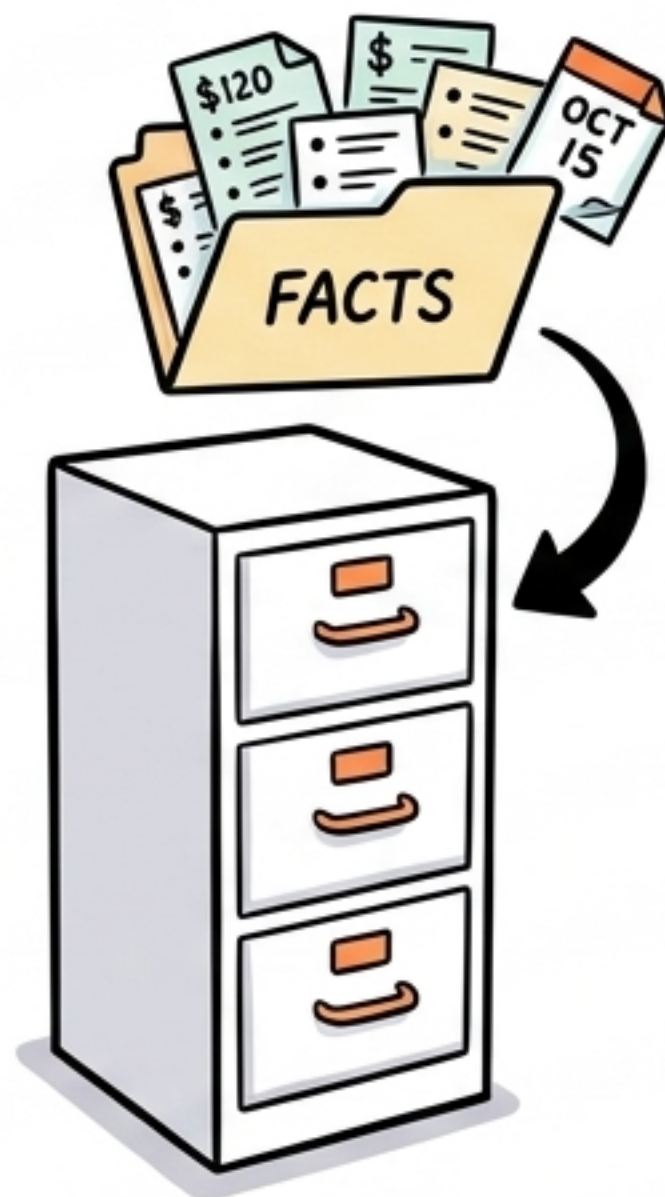
D5: ANTI-BLOAT & MEMORY

THE TRASH CAN



TRIM VERBOSE TOOL RESULTS TO ONLY RELEVANT FIELDS BEFORE THEY ENTER THE CONTEXT WINDOW.

THE FILING CABINET



EXTRACT PRECISE TRANSACTIONAL FACTS (AMOUNTS, DATES) INTO A PERSISTENT CASE-FACTS SCRATCHPAD OUTSIDE THE SUMMARIZED HISTORY.

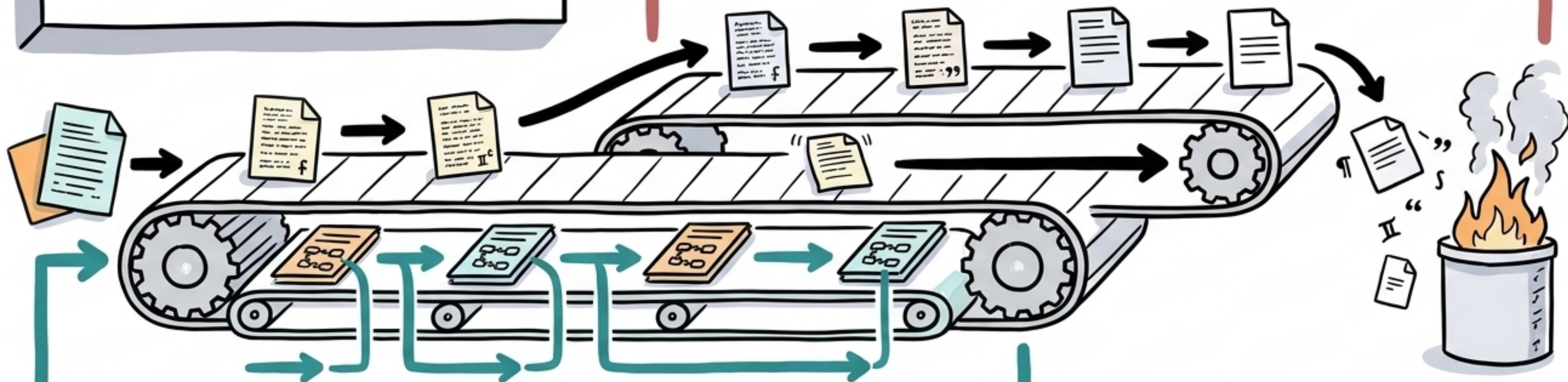
THE RELIEF VALVE



USE MID-SESSION TO COMPRESS VERBOSE DISCOVERY OUTPUT AND RESCUE A DEGRADING CONVERSATION.

D5: RELIABILITY & PROVENANCE

SUMMARIZATION DESTROYS ATTRIBUTION.



PROVENANCE MUST BE STRUCTURAL.

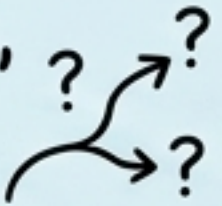
REQUIRE SUBAGENTS TO OUTPUT STRUCTURED CLAIM-SOURCE MAPPINGS. BIND SOURCES TO CLAIMS AS DATA, PRESERVED THROUGH EVERY SYNTHESIS HOP.



HANDLING CONFLICTS:

WHEN SOURCES DISAGREE, DON'T LET THE MODEL SILENTLY AVERAGE THEM! ANNOTATE THE CONFLICT, CARRY PUBLICATION DATES, AND LET THE COORDINATOR RECONCILE.

ARCHITECT DIAGNOSTIC: WORKFLOWS

TASK PROFILE: OPEN-ENDED / UNPREDICTABLE (Legacy exploration, Incident response)	TASK PROFILE: PREDICTABLE / FIXED (Standard 5-aspect Code Review)
STRATEGY: DYNAMIC DECOMPOSITION   Map the terrain, prioritize, let the plan evolve. 	 STRATEGY: PROMPT CHAINING  Fixed, reproducible steps that are easy to regression-test. 

EXECUTION VS PLAN MODE



- SINGLE-FILE BUG FIX?
→ DIRECT EXECUTION.



- 45-FILE MIGRATION WITH
COMPETING APPROACHES?
→ PLAN MODE.



THE CCAR-F ANTI-PATTERN CHEAT SHEET



RELYING ON TEMPERATURE
0 TO ENFORCE
BUSINESS POLICY.



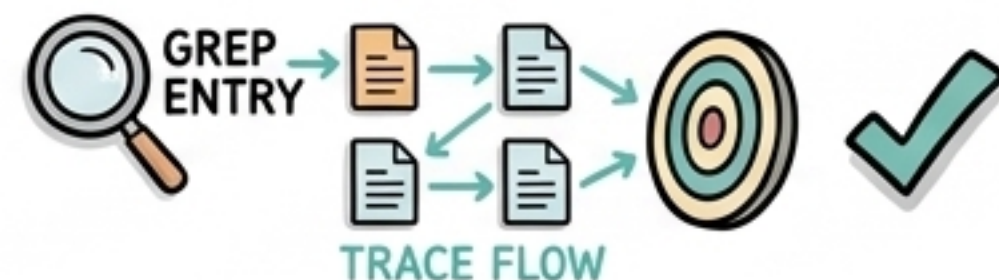
USING INTERCEPTION
HOOKS TO BLOCK
VIOLATING TOOL CALLS.



READING LEGACY
FILES ALPHABETICALLY
UNTIL CONTEXT FILLS.



GREP FOR ENTRY
POINTS, THEN READ
INCREMENTALLY
TO TRACE FLOWS.



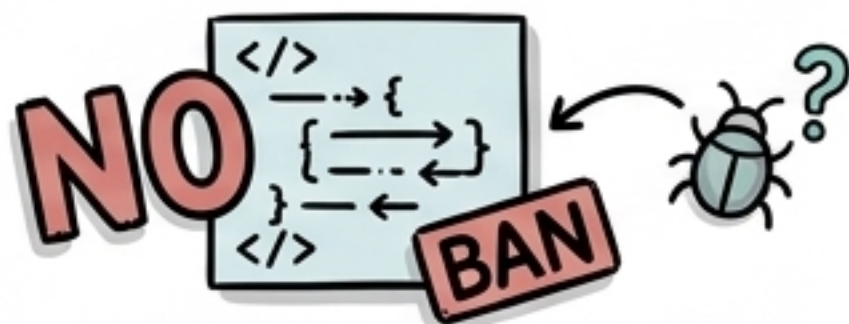
RESUMING A SESSION
WHEN TARGET FILES
HAVE BEEN HEAVILY
MODIFIED.



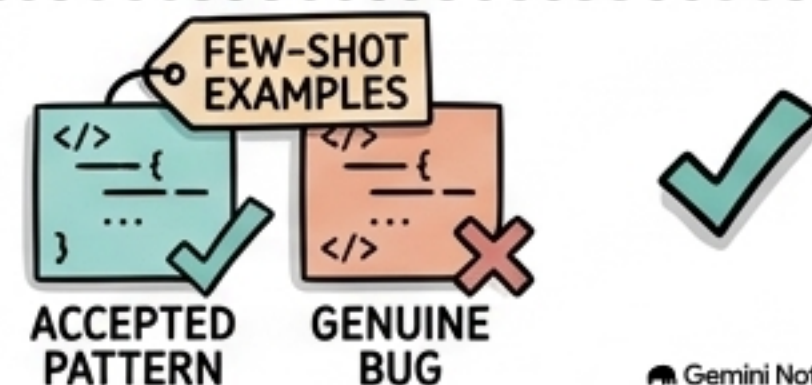
START FRESH WITH AN
INJECTED SUMMARY
(DISCARD STALE TOOL
RESULTS).



BANNING IDIOMATIC
CODE PATTERNS TO
STOP FALSE POSITIVES.



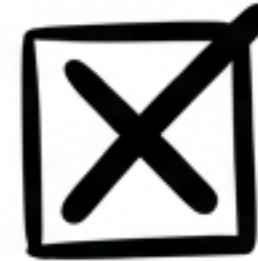
SUPPLYING FEW-SHOT
EXAMPLES CONTRASTING
ACCEPTED PATTERNS VS
GENUINE BUGS.



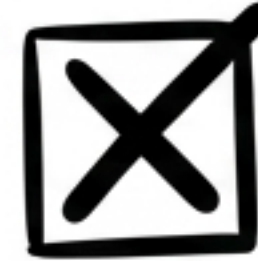
READY FOR THE EXAM



HANDS-ON EXPERIENCE
WITH THE CLAUDE AGENT
SDK & CLAUDE CODE.



REVIEWED THE OFFICIAL
ANTHROPIC MCP
DOCUMENTATION.



MASTERED THESE
SKETCHNOTES.

YOU GOT THIS.

GOOD LUCK ON THE CCAP-E